

Wireless EV Charging System

Design and Simulation with Power Factor Correction

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Abstract

This report presents the design and simulation of a static wireless charging system for electric vehicles. The system covers the full power chain, from the AC grid to a Li-ion battery, including an AC/DC rectifier, a high-frequency inverter, transmitter and receiver coils, a secondary rectifier and a compensation network. Three compensation topologies are compared: series-series, series-parallel and LCC/L. Two power factor correction strategies are implemented and compared: a passive LC filter and an active boost converter with a dual PI control loop. Results show that the choice of compensation network affects efficiency more than the choice of PFC method, and that active PFC trades efficiency for a near-unity power factor and low harmonic distortion.

1 Introduction

Wireless charging removes the physical connector between a vehicle and the grid. It also removes the wear, weather exposure and alignment problems that come with cables. For this to be practical, the system must transfer power efficiently and draw a clean current from the grid.

The global transition toward sustainable mobility has placed electric vehicles at the centre of energy strategies. The number of EVs on the road is expected to exceed 26 million by 2030, and the wireless charging market is projected to reach 1.2 billion USD over the same period. This growth creates a real engineering challenge: making wireless charging both efficient and grid-friendly.

This project addresses that challenge through simulation. It designs a complete static wireless charging system, compares multiple compensation topologies and evaluates two strategies for power factor correction.

Key Insight

Wireless charging is not just about transferring power without cables. It is about transferring power without degrading the grid that supplies it.

2 System Architecture

The system is divided into two sides: the charger side, installed on the ground, and the vehicle side, embedded in the car.

On the charger side, the AC grid voltage is rectified by a diode bridge, stabilised by a DC-link filter and then converted to high-frequency AC by an inverter. The high-frequency AC feeds the primary coil.

On the vehicle side, the secondary coil captures the magnetic field. Its output is rectified and used to charge the battery. Voltage, current and state of charge are monitored.

A compensation network is placed on both sides. Its role is to cancel the reactive part of the impedance so that the system runs near resonance. Without compensation, reactive power dominates and efficiency collapses.

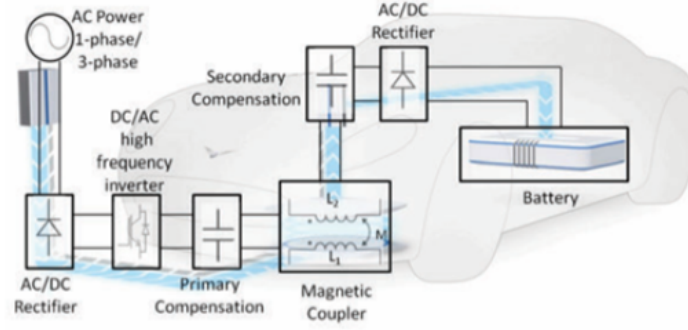


Figure 1: Complete power chain of the wireless charging system, from the three-phase grid connection on the left to the onboard battery on the right.

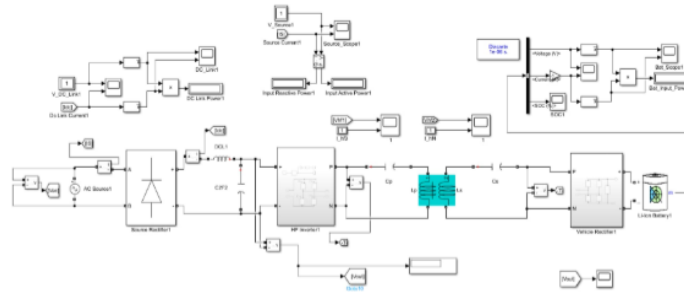


Figure 2: The Simulink model implementing the full system, including the primary inverter, the resonant coupling, the secondary rectifier and the battery load.

3 Compensation Topologies

Compensation networks come in two families: basic topologies and hybrid topologies.

Basic topologies place a capacitor in series or in parallel with each coil. Four combinations exist: series-series (SS), series-parallel (SP), parallel-series (PS) and parallel-parallel (PP). Among these, SS is the most widely used for EV charging because of its simplicity and its load-independent resonant frequency.

Hybrid topologies combine inductors and capacitors to improve behaviour under load variation or misalignment. Common examples include LCC/LCC, CCL/LC, LC/LC and LCC/LCCL.

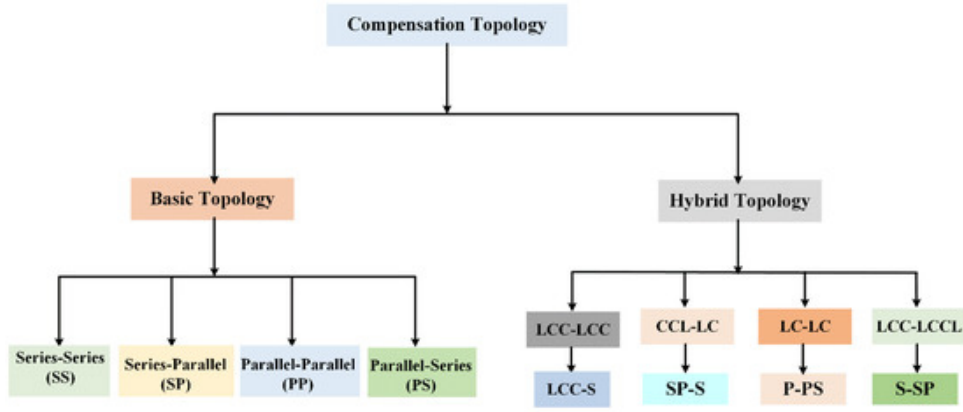


Figure 3: Overview of the two compensation families: basic topologies on the left, hybrid topologies on the right.

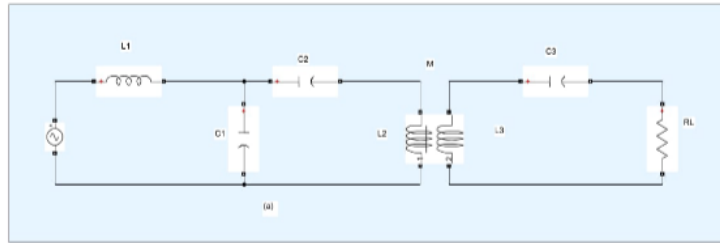


Figure 4: Equivalent circuit of the hybrid LCC-S network, showing the two compensation inductors and the two capacitors on the primary side.

Three configurations were selected for detailed simulation: SS, SP and LCC/L. The first two represent the basic family, the third represents the hybrid family.

4 Power Factor Correction

High-frequency converters and inductive coupling generate reactive power and harmonics. Without correction, the power factor drops, line losses increase and the system violates grid standards.

Two strategies were implemented.

4.1 Passive PFC

A simple LC filter tuned to a low frequency, around 290 Hz in this project. The filter reduces the phase shift between voltage and current. It is cheap, robust and easy to integrate. Its main weakness is that it does not actively shape the current waveform, so total harmonic distortion remains high.

4.2 Active PFC

A boost converter with a dual PI control loop. The outer loop regulates the DC bus voltage. The inner loop shapes the inductor current to follow a sinusoidal reference in phase with the grid voltage. The result is a near-unity power factor and very low distortion.

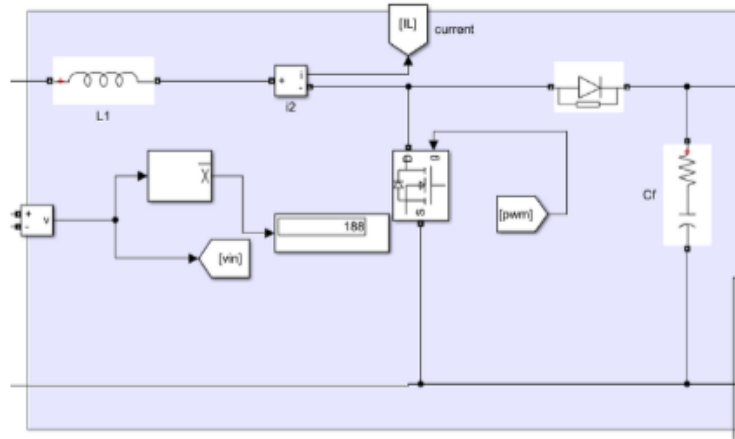


Figure 5: Boost converter used as the active PFC stage, showing the inductor, the switch and the output capacitor.

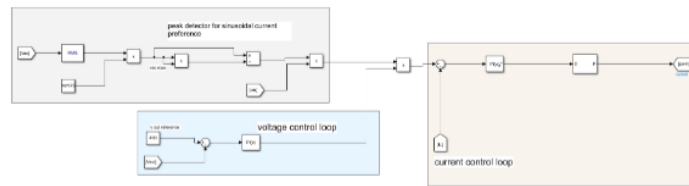


Figure 6: Control architecture of the boost PFC: peak detector, sinusoidal reference generation, and the nested voltage and current PI loops driving the PWM.

Why the Boost Converter

Among the possible topologies for active PFC, the boost converter is the most common. It steps up the voltage to the DC bus level, and it keeps the input current continuous, which makes the filter design simpler. The buck converter cannot reach the required DC bus voltage, and the buck-boost converter stresses the switch more.

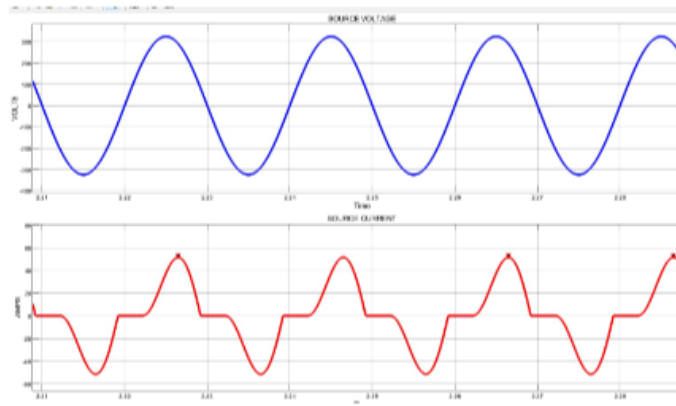
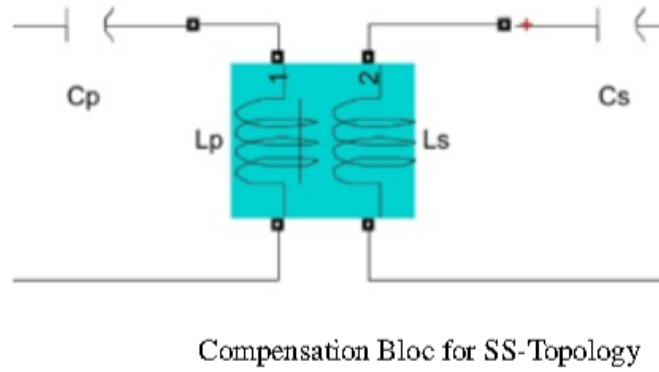
5 Simulation Results

Five configurations were simulated and evaluated on three metrics: efficiency, power factor and total harmonic distortion.

Table 1: Performance of each configuration.

Configuration	Efficiency	Power Factor	THD
SS topology, passive PFC	84%	0.9256	43%
SP topology, passive PFC	72%	0.9471	43%
LCC/L topology, passive PFC	97%	0.9991	46.99%
SP topology, active PFC	75%	0.9886	15.86%
SS topology, active PFC	78%	0.9998	2.99%

5.1 Passive PFC with SS Topology

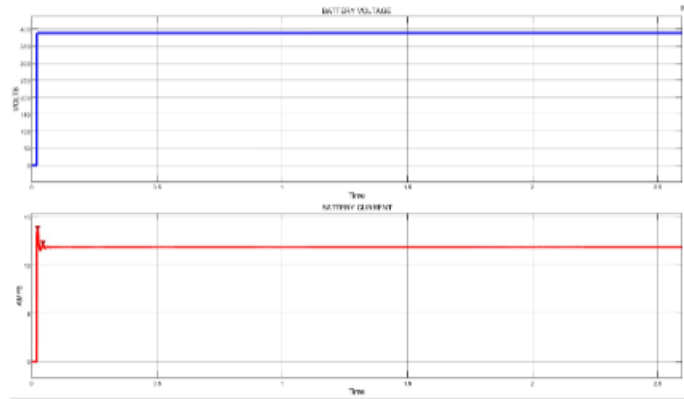


Source voltage and Current

Figure 7: Series-series compensation block used in the SS configuration plus Source voltage and current waveforms with the SS topology and passive PFC.

Table 3.2: Performance metrics of SS topology with passive PFC

Metric	Value
Efficiency	84 %
Power Factor (PF)	0.9256
Total Harmonic Distortion (THD)	43 %

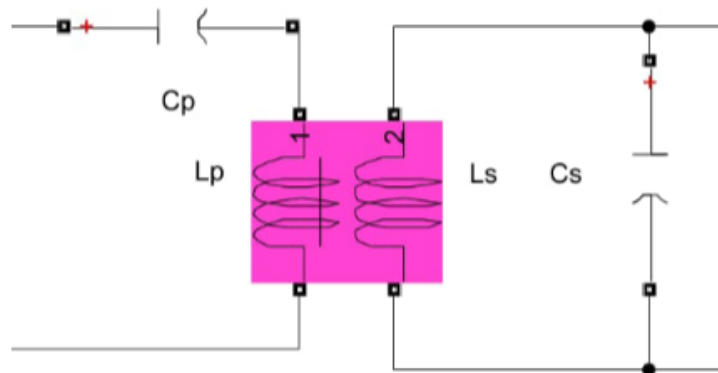


battery characteristics

Figure 8: Battery voltage and current during charging, SS topology with passive PFC.

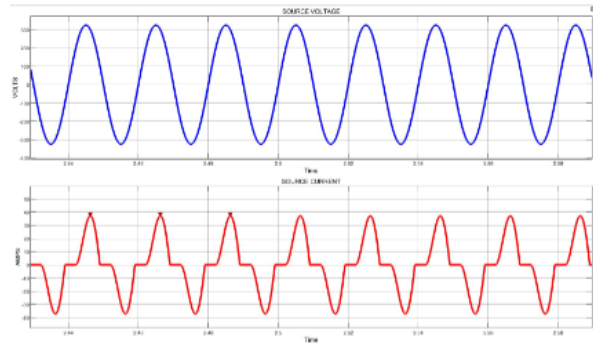
With the SS topology, the passive filter brings the power factor to 0.9256 while keeping efficiency at 84 percent. Distortion remains at 43 percent.

5.2 Passive PFC with SP Topology



Compensation Bloc for SP-Approach

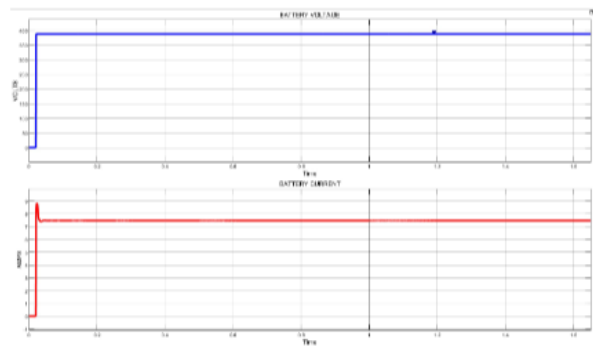
Figure 9: Series-parallel compensation block used in the SP configuration.



Source voltage and Current

Table 3.3: Performance metrics of Sp Approach with passive PFC

Metric	Value
Efficiency	72 %
Power Factor (PF)	0.9471
Total Harmonic Distortion (THD)	43 %

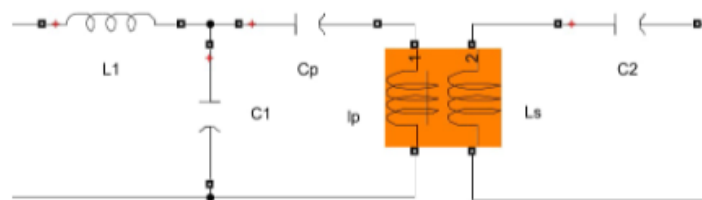


Battery characteristics

Figure 10: Source voltage and current, and battery characteristics, for the SP topology with passive PFC.

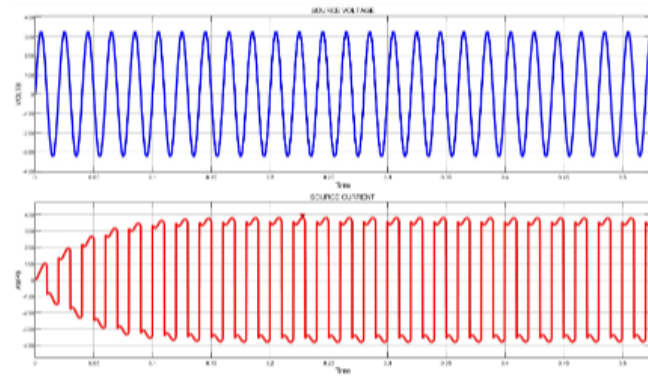
With the SP topology, efficiency drops to 72 percent. The power factor improves slightly to 0.9471, but the losses are significant.

5.3 Passive PFC with LCC/L Topology



Compensation Bloc for SP-Approach

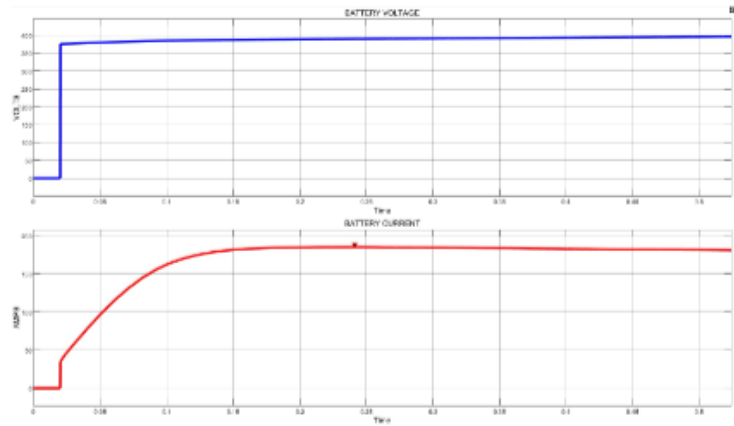
Figure 11: Hybrid LCC/L compensation block, combining an inductor on the primary and a capacitor on the secondary side.



Source voltage and Current

Table 3.4: Performance metrics of LCC/L Approach with passive PFC

Metric	Value
Efficiency	97 %
Power Factor (PF)	0.9991
Total Harmonic Distortion (THD)	46.99 %

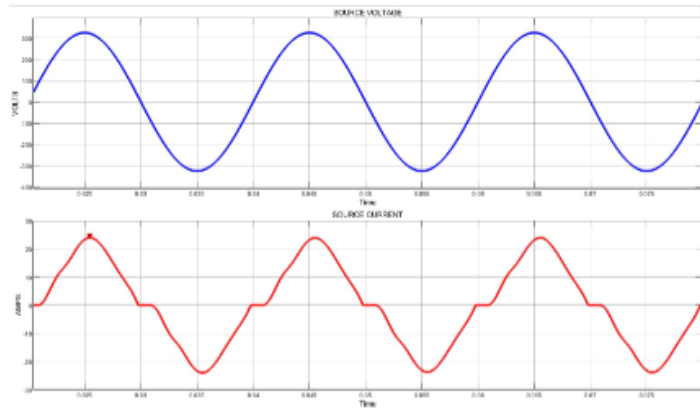
Figure 12: Source voltage and current waveforms with the LCC/L topology and passive PFC.

Battery characteristics

Figure 13: Battery voltage and current during charging, LCC/L topology with passive PFC.

With the LCC/L topology, the results change completely. Efficiency reaches 97 percent and the power factor reaches 0.9991. This shows that a well-chosen compensation network can achieve near-unity power factor without active correction.

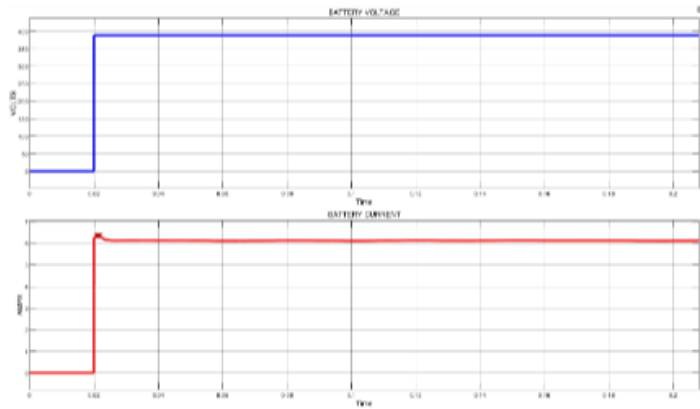
5.4 Active PFC with SP Topology



Source Voltage and Source Current

Table 3.5: Performance Results

Parameter	Value
PF	0.9886
THD	0.1586
Efficiency (η)	75%

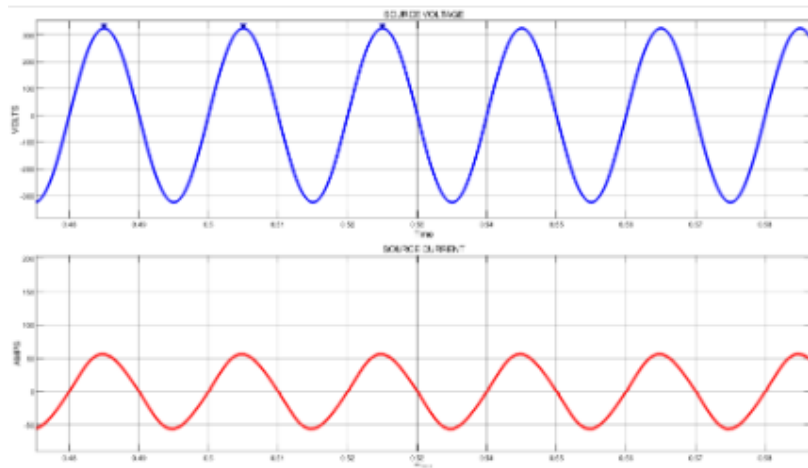


Voltage and Current Battery

Figure 14: Source voltage and current together with battery characteristics, SP topology and active PFC.

With the SP topology and the boost converter, the power factor rises to 0.9886 and THD drops below 16 percent. Efficiency remains at 75 percent because of switching losses.

5.5 Active PFC with SS Topology

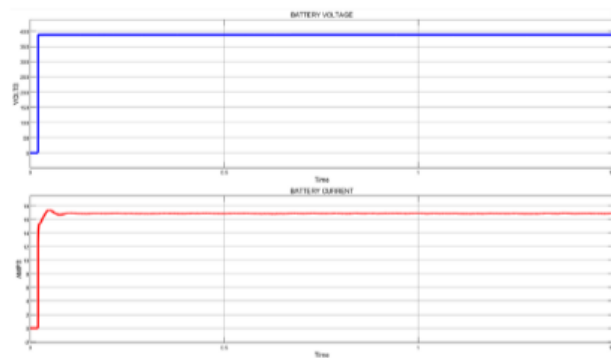


Voltage source and current source

Figure 15: Source voltage and current waveforms with the SS topology and active PFC.

Table 3.6: Performance metrics of SS Approach with active PFC

Metric	Value
Efficiency	78 %
Power Factor (PF)	0.9998
Total Harmonic Distortion (THD)	2.996 %



Battery voltage and Battery current

Figure 16: Battery voltage and current during charging, SS topology with active PFC.

With the SS topology and the boost converter, the results are the best of the study. Power factor reaches 0.9998 and THD drops to 2.99 percent, which meets grid standards. Efficiency is 78 percent.

Table 3.7: Comparative Analysis of PFC Topologies

Parameter	LCC (Passive)	SP (Passive)	SS (Active)	SP (Active)
Components	L, C, C (3 elements)	C, C (2 elements)	Boost Converter + PI Control	Boost Converter + PI Control
Efficiency (%)	75-85	70-80	92-96	90-94
Power Factor	0.90-0.95	0.85-0.92	0.998-0.999	0.98-0.99
THD (%)	15-30	20-35	<3	<5
Cost	Low	Lowest	High	Medium
Complexity	Simple	Simple	Complex	Moderate
Frequency	290 Hz	300 Hz	30 kHz	30 kHz
Best For	Low-power non-critical apps	Basic low-cost systems	High-performance EV charging	General-purpose power supplies

Figure 17: Side-by-side comparison of the four PFC topologies on efficiency, power factor and harmonic distortion.

6 Discussion

The results reveal a clear trade-off.

Passive PFC keeps the circuit simple and the cost low. The LCC/L topology with passive PFC reaches 97 percent efficiency and a power factor of 0.9991, which is remarkable for a filter-only solution.

Active PFC produces the cleanest current. The SS topology with active PFC reaches a power factor of 0.9998 and reduces THD to below 3 percent, meeting the strictest grid standards. Efficiency is lower because of switching losses.

The compensation topology has more influence on efficiency than the PFC method. The LCC/L configuration, for example, outperforms every other configuration on efficiency regardless of the PFC method used.

Important Limitations

- The battery is modelled as a resistive load. Real battery behaviour includes electrochemical dynamics that are not captured here.
- Semiconductor switches are treated as ideal. Switching losses are approximated but not fully simulated.
- Coil misalignment is not studied. In real conditions, misalignment is the main cause of efficiency loss.
- Thermal behaviour and electromagnetic interference are not simulated.

7 Conclusion

This project demonstrates that a well-designed wireless charging system can deliver high efficiency and clean power quality at the same time. The choice of compensation topology is the primary lever for efficiency. The choice of PFC method is the primary lever for power quality. Together, they define the operating point of the system.

The results also show that a passive LC filter can achieve excellent power factor when combined with the right compensation network, which makes wireless charging viable even for cost-sensitive applications.

8 Tools and Environment

Table 2: Tools used in this project.

Category	Tool	Role in the project
Simulation	MATLAB/Simulink	Full system modelling and simulation
Power library	Simscape Electrical	Power electronics components and circuits
Control	PI control blocks	Voltage and current regulation for active PFC
Analysis	Scope and measurement blocks	PF, THD and efficiency calculation

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